

IoT-Based Smart Energy Monitoring System

Academic / Portfolio Project Design Report

1. Abstract

This project presents the design of an IoT-Based Smart Energy Monitoring System for monitoring electrical parameters such as voltage, current, power, and energy consumption. A microcontroller receives data from suitable sensors, calculates the electrical quantities, and can transmit the information to an IoT platform for remote monitoring. The proposed system is intended as an academic and portfolio project design that demonstrates the integration of electrical measurement, embedded systems, and Internet of Things technology.

2. Introduction

Energy monitoring is important in residential, commercial, and industrial electrical systems. Conventional energy meters generally provide local readings, while an IoT-enabled monitoring system can make energy information available remotely. This project focuses on a simple architecture for sensing electrical parameters, processing them using a microcontroller, and displaying or transmitting the calculated values.

3. Problem Statement

Users may not have convenient access to real-time information about their electrical energy consumption. A smart monitoring system can provide better visibility of electrical parameters and help users understand how energy is being used.

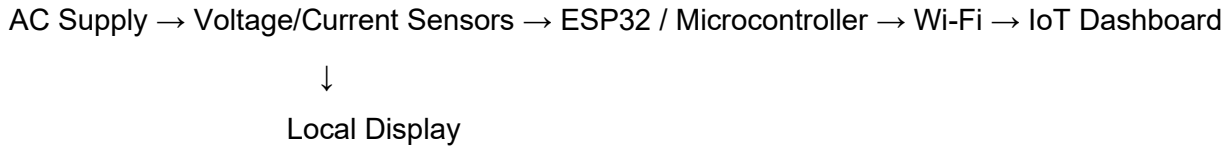
4. Objectives

- To monitor voltage and current.
- To calculate electrical power and energy consumption.
- To use a microcontroller for data processing.
- To demonstrate IoT-based remote monitoring.
- To provide a simple platform for studying smart energy management.

5. Proposed System

The proposed system uses voltage and current sensing circuits connected to a microcontroller such as an ESP32. The controller processes the sensor values and calculates the required electrical parameters. Through Wi-Fi connectivity, the measured data can be transmitted to a suitable IoT dashboard or cloud service for visualization.

6. Block Diagram



7. Main Components

- ESP32 or similar microcontroller
- Voltage sensing module
- Current sensing module
- Display module (optional)
- Wi-Fi network
- IoT dashboard / cloud platform
- Regulated power supply
- Electrical load
- Connecting wires and supporting components

8. Working Principle

1. The electrical load is connected to the monitored supply.
2. Voltage and current sensors obtain electrical measurements.
3. The sensor outputs are provided to the microcontroller.
4. The controller processes the readings and calculates electrical quantities.
5. Energy consumption is estimated by accumulating power with respect to time.
6. The processed values can be displayed locally.
7. Through Wi-Fi, the values can also be sent to an IoT platform for remote monitoring.

9. Basic Calculations

For a simplified AC load model:

Apparent Power, $S = V \times I$

For an AC system where power factor is known:

Real Power, $P = V \times I \times PF$

Electrical Energy = Power × Time

In a practical implementation, accurate real-power and power-factor measurement requires appropriate sampling, calibration, and consideration of the phase relationship between voltage and current.

10. IoT Architecture

The ESP32 can connect to a Wi-Fi network and transmit processed measurements to an IoT service. A dashboard can display parameters such as voltage, current, power, and accumulated energy. The communication method and cloud platform can be selected according to the implementation requirements.

11. Software / Programming

The embedded program can perform sensor acquisition, calibration, calculation, display updates, Wi-Fi communication, and data transmission. The control flow can be organized as initialization → sensor reading → calculation → display/upload → delay → repeat.

12. Expected Results

The proposed design is expected to provide periodic readings of voltage, current, power, and energy consumption and make them available for local or remote viewing. The actual accuracy, update rate, and communication reliability depend on the selected sensors, calibration, load, network, and implementation.

13. Advantages

- Remote monitoring capability
- Real-time visibility of energy usage
- Integration of EEE and embedded-system concepts
- Expandable for additional sensors and control features
- Useful for educational demonstrations and portfolio work

14. Limitations

- Measurement accuracy depends on sensor quality and calibration.
- IoT features depend on network availability.
- Practical AC mains implementation requires proper electrical isolation and safety measures.
- The design requires hardware testing before being treated as a validated measurement system.

15. Applications

- Residential energy monitoring

- Laboratory demonstrations
- Small commercial monitoring systems
- Educational smart-grid experiments
- Energy-awareness and data-logging projects

16. Future Scope

- Mobile dashboard and notifications
- Historical energy-consumption graphs
- Overload alerts
- Power-factor monitoring
- Automated load control
- Integration with renewable-energy systems
- Machine-learning-based consumption analysis

17. Conclusion

The IoT-Based Smart Energy Monitoring System combines electrical measurement, embedded processing, and wireless communication in a single academic project design. It provides a useful foundation for understanding smart metering and energy-management concepts. A future physical prototype can be used to calibrate the sensors and validate the system under real operating conditions.

18. Project Status

This document describes an Academic / Portfolio Project Design. It does not claim that a physical prototype has been built or experimentally validated. Any numerical or performance results should be added only after actual implementation and testing.

19. Suggested GitHub Structure

IoT-Smart-Energy-Monitoring/

```
|— README.md
|— Report/
|   |— IoT_Based_Smart_Energy_Monitoring_System_Report.pdf
|— Design/
|   |— block_diagram.png
|— Code/
|   |— energy_monitor.ino
```

20. References

ESP32 technical documentation and microcontroller reference material.

Standard electrical measurements and energy-metering textbooks.

Sensor datasheets selected during physical implementation.

IoT platform documentation selected during physical implementation.